Day 5

APTITUDE

Block-1

1) A train travels 60 km in 1 hour. Find speed in m/s.

Sol:

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

km/h → m/s

m/s → km/h

⇒ 18/5

⇒ 5/18

60×5

300

$60 \times 5/18$

$\Rightarrow \frac{300}{18} \Rightarrow \frac{150}{9}$

$\boxed{\Rightarrow 16.67}$ → Answer

9 | 150 ⇒ 16.67
144
60
54
6

2) A car travels at 40 km/hr for 2 hours and 60 km/hr for next 2 hrs. Find average speed.

Sol: Given

40 km/hr for 2 hrs → Distance = 40 × 2 → 80 km

60 km/hr for 2 hrs → Distance ⇒ 60 × 2 ⇒ 120 km

Total Distance ⇒ 80 + 120 → 200 km

Total Time ⇒ 2 + 2 ⇒ 4 hrs

Average speed ⇒ Total Distance / Total Time

$\Rightarrow \frac{200}{4} \Rightarrow 50$ km/hr

$\boxed{\Rightarrow 50 \text{ km/hr}}$ → Answer.

3) Two trains moving in opposite direction at 50 km/h and 70km/h. Find relative speed.

Sol:

Given

Two trains opposite direction

Relative speed = sum of speeds

$= 50 + 70$

$\Rightarrow 120 km/hr$

$\boxed{120 km/hr}$ → Answer.

4) A person walks at 5km/h and runs at 10km/h. Find Time taken to cover 15 km if half distance is walked and half distance is run.

Sol:

Given

Total Distance = 15 km

Half walked $\Rightarrow 15/2 \Rightarrow 7.5$ km at 5km/h

Half run $\Rightarrow 15/2 \Rightarrow 7.5$ m at 10km/h

Time = Distance/speed

walking Time $\Rightarrow \frac{7.5}{5} \Rightarrow 1.5 hr$

Running Total Time

$\Rightarrow \frac{1.5}{10} \Rightarrow 0.15 hr$

Total Time

$\Rightarrow 1.5 + 0.15$

$\Rightarrow 2.25 hr$

$\boxed{\text{Answer : 2.25 hours}}$

1.50
0.15
2.25

5. If a train crosses a pole in 10 seconds and its length is 100 meters, find its speed.

Sol:

Given

Train crosses pole → Distance

→ Length of train = 100m

→ Speed → $\frac{100}{10}$ → 10m/s

Convert to km/h

m/s → km/h → 18/5 $10 \times 18/5$

→ $\frac{180}{5}$

→ 36 km/h

Block - 2 → VERBAL ABILITY

1) why are algorithms are important?

Algorithms are important, Because, it plays a crucial role in optimizing system performance.

2) what happens if inefficient algorithms are used?

If use inefficient algorithm means it take more time to execute and reduce the efficiency speed.

3) what is optimization in compiling?

optimization means it improves performance by reducing execution time.

Grammar correction.

1) He have been working here since 2 years.

Specific time we don't use here

He has been working here for 2 years.

2) She did not wrote the code.

She did not write the code.

3) They is preparing for interviews.

They are preparing for interviews.

Block 3 → COMMUNICATION PRACTICE

→ Explain a recent technical topic you learned.

Recently, I have learned problem solving, DSA, DBMS and OS concepts. These are very important topics to crack a product-based company interview.

I started from basics and used java to solve problems. Then I learning Data structures to organize data efficiently. And, OS helps in managing memory and system resours. DBMS is us to store the data and retrieve the data.